

RAUSCHEN

2025 | emergent media system | Processing | OpenGL

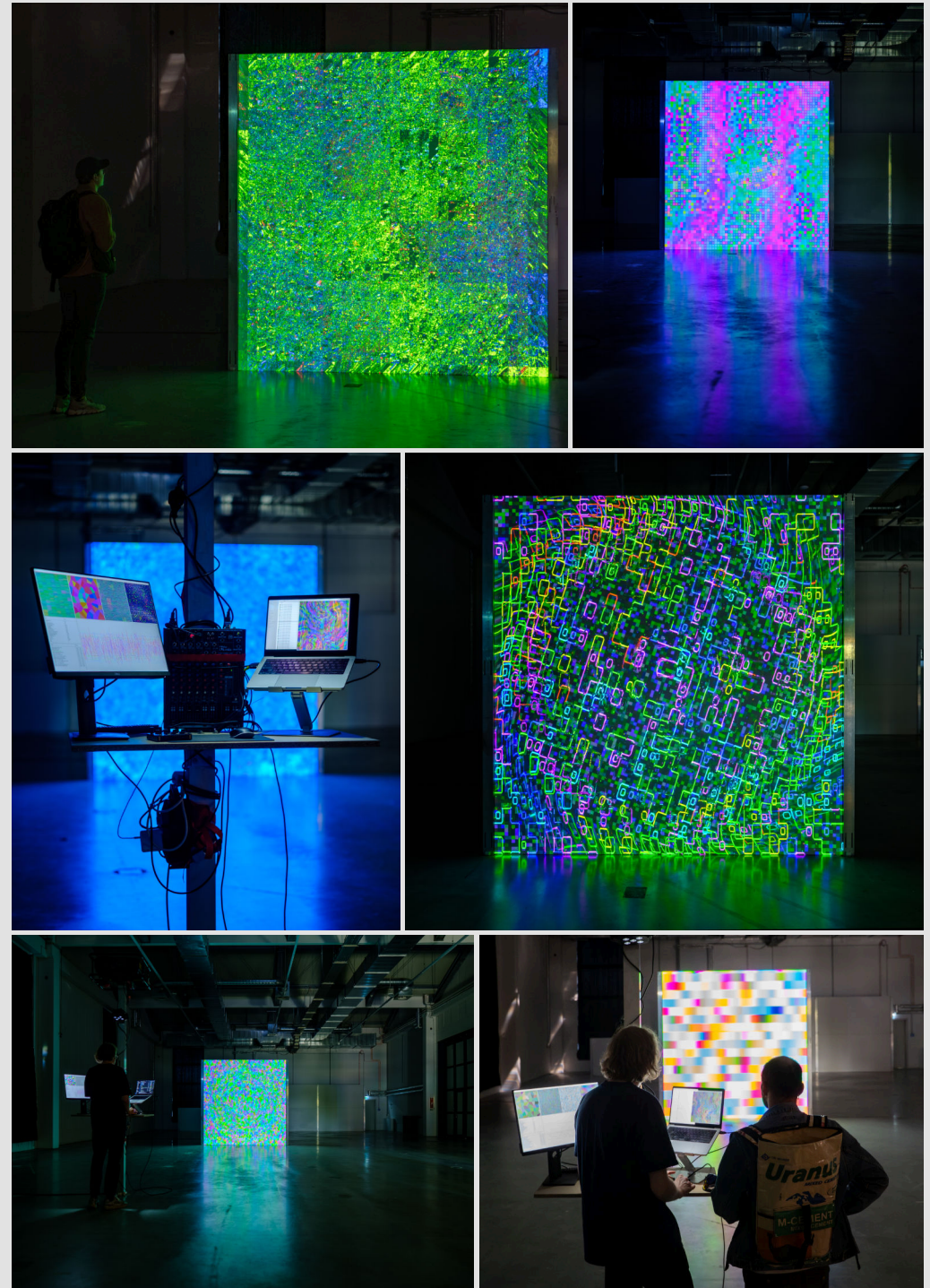
Every digital image is a limited, but unimaginably large space of possibilities. All these possibilities together are its probability space. In a regular 24-bit image file, each pixel is defined by 256 shades each of red, green and blue, which comes out to 16,777,216 possible colours. For a one megapixel image (1,000 pixels wide and 1,000 pixels tall) there are $16,777,216^{1,000,000}$ permutations. That number far exceeds how many atoms there (probably) are in the observable universe (10^{80}). Actually it is $7.87 \times 10^{7,224,639}$ times larger. Even if every atom in the universe was another universe full of atoms that are universes full of atoms and so on, we would have to go 90,309 levels deep to finally have as many atoms as there are permutations of a 24-bit megapixel image.

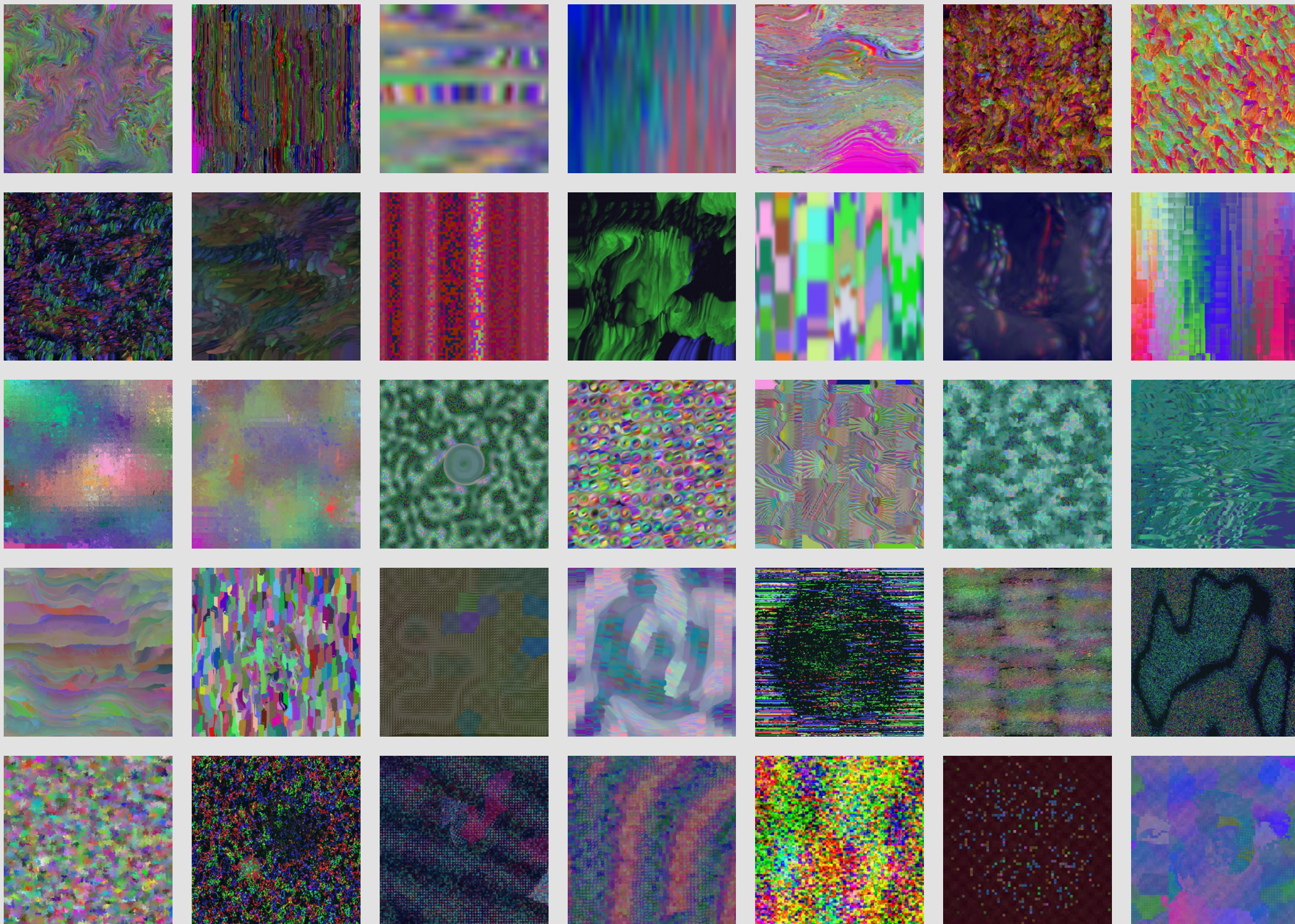
Most of it is just meaningless noise. But how do we find the infinitely small rest of it? You know, the rest that contains (among other things) all photos, drawings or graphics ever made, will ever be made or could ever be made? Usually, we go at it from the other direction: We take a picture. We make a drawing. We create graphics. Yet we are always limited by the fact that meaning needs to exist first, either in the world or in our minds, to be found within the noise.

Or we could just leave it up to chance.

RAUSCHEN ("noise") is a real-time emergent media system exploring the probability space of a 1,000x1,000 image. It generates textures with a variety of techniques such as white or Perlin noise and recursively feeds them into a modular palette of per-pixel algorithms. They each represent visual effects that, running in succession, synthesise exponentially random patterns. While contemporary image generation models try to impose meaning onto the noise based on their training data, RAUSCHEN aims to chart new areas hidden within the noise itself. In the end, a human decides what is worth keeping.

How can we find meaning within all this noise?





tech rider // projection

- **exhibition space:**

- dark environment
- sufficient viewing space in front of projection, depending on its size (3x screen width)

- **projector:**

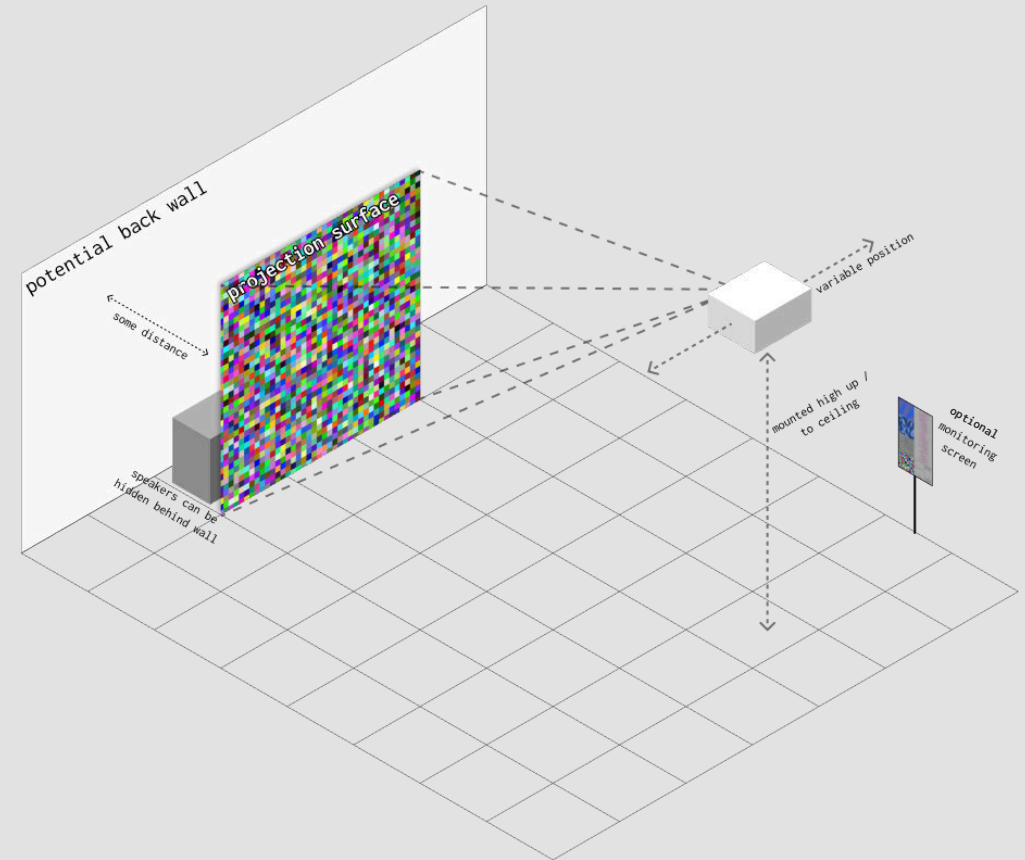
- ideally a laser projector to mitigate light bleed
- depending on ambient light situation (>4000 ANSI Lumen, more is better)
- minimum 1000x1000p resolution
- >60Hz
- keystone correction

- **projection surface:**

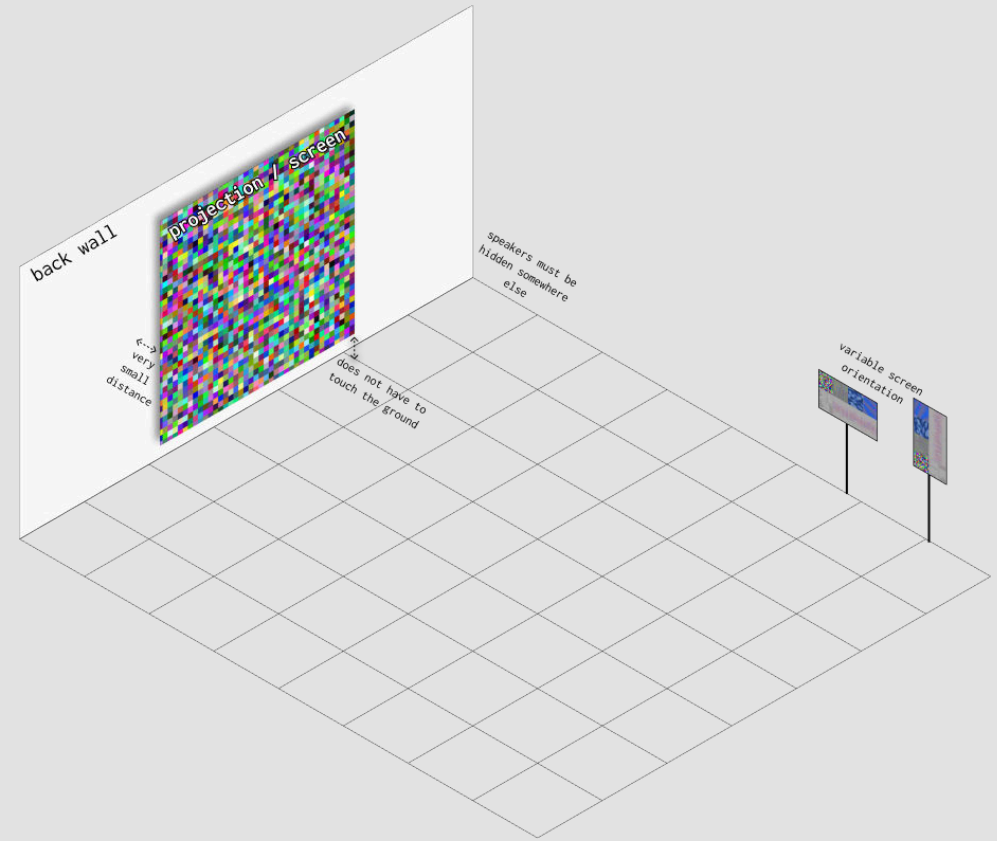
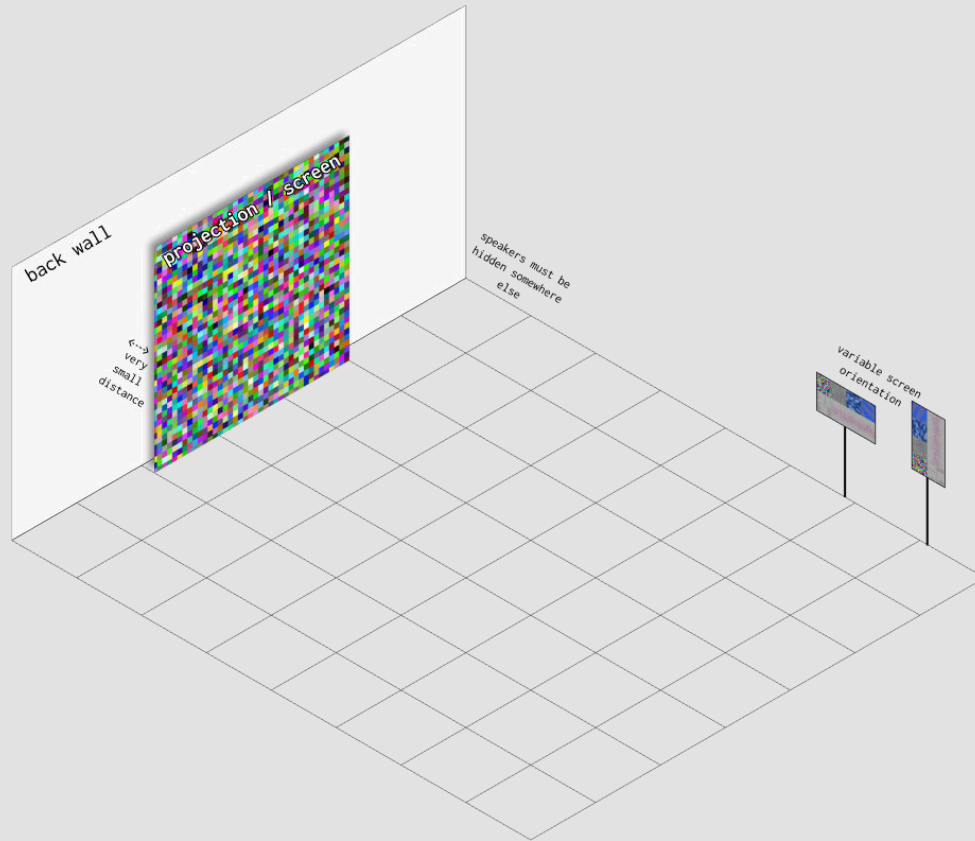
- square, at least 2x2m, bigger is better
- white plane without imperfections (white paint usually does it)
- should be clearly separated from wall
- separation distance may be small (>5cm), but should ideally be larger (>1m)

- **projector mounting:**

- projector should be mounted from ceiling or a high up stand, to avoid visitor shadows on projection surface
- actual position in relation to surface is variable, as long as at least 1000x1000px of the (keystoned) projection land on the projection surface



tech rider // alternate projection surface configs



tech rider // other info

- **sound system:**

- sufficient for exhibition space, needs at least some low frequencies / bass
- accessible via AUX
- ideally hidden behind screen if enough space

- **optional monitoring screen:**

- >2560x1440p display with HDMI / Thunderbolt, bigger is better
- wall mounted or on a stand
- (the table with controls from the pictures is not needed for an exhibition context)

- **computer:**

- M1 Pro or upwards MacBook Pro / Mac Mini

- **setup:**

- projection surface + projector / LED wall and sound system can be set up in advance
- installation setup (video mapping, controls and computer setup) must be done by the artist

if you have any other ideas how to set RAUSCHEN up, please do not hesitate to get in contact @ lucca.vitters@gmail.com

(LED wall?)

- **exhibition space:**

- dark-ish environment
- sufficient viewing space in front of screen, depending on its size (3x screen width)

- **LED wall:**

- square, at least 2x2m, bigger is better
- minimum 1000x1000p resolution, ideally 1:1 pixel mapping
- >120Hz + fast pixel/LED response time

disclaimer: has not yet been tested on an LED wall

